Nebius Object Storage
System-Under-Test Description

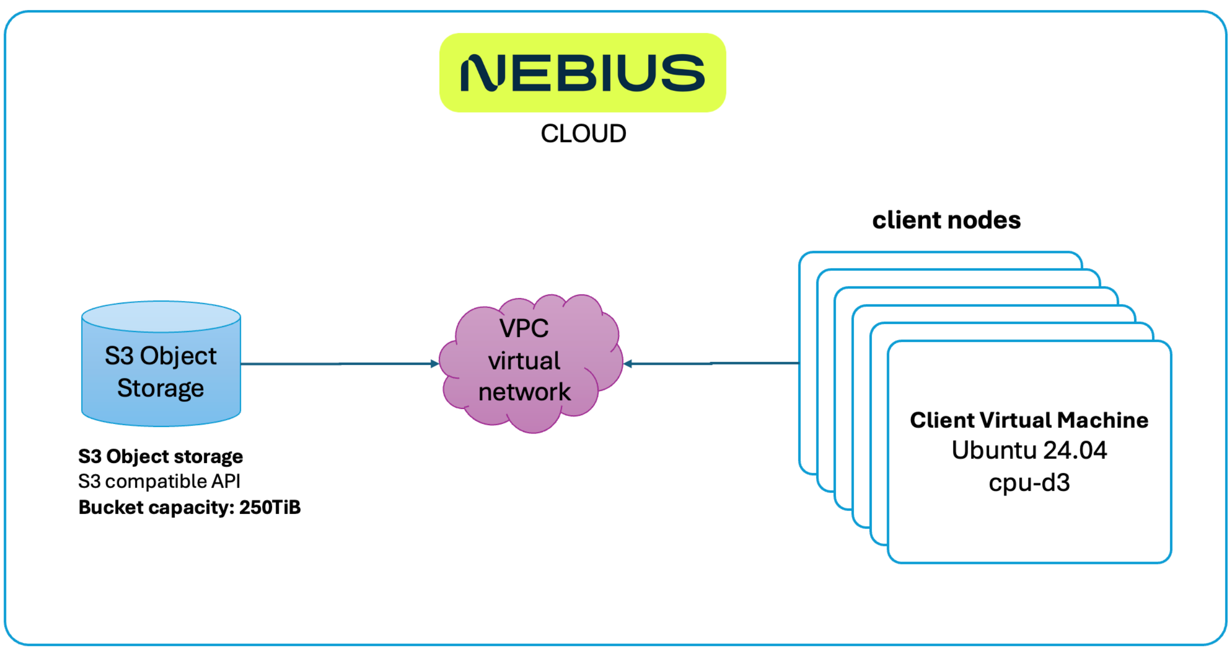
System name: Nebius Object Storage
MLPerf Storage division: CLOSED
Deployment: Cloud

Overview

Nebius AI Cloud is a specialized, AI-native cloud platform. It is built specifically to provide high-performance GPU infrastructure and full-stack software solutions for training, fine-tuning, and deploying generative AI, robotics, and large-scale machine learning models.

Nebius Object storage

Nebius Object Storage is a fully S3-compatible object storage service designed specifically for AI workloads. Storage class used for tests: Enhanced Throughput designed specifically for AI Training & Checkpointing use cases.



Cloud test LAB diagram

Architecture

Property	Value
Storage location	Remote network-attached, separate from compute nodes
Benchmark access protocol	S3 API
Product access protocol	S3 API
Client-side software	open-source mlpstorage python package
Total capacity	250 TiB (variable)

Capabilities

Capability	Supported
Multi-host access	Yes
Simultaneous writes from multiple hosts	Yes
Simultaneous reads from multiple hosts	Yes
Remapping time	not applicable always simultaneously read/write capable

Client Node Configuration

Client node based on publicly available Nebius CPU-only virtual machine preset:
Non-GPU AMD EPYC Genoa (cpu-d3): **64vcpu-256gb RAM**.

And limited to OS available RAM by using following cloud-init profile:

```
write_files:
  - path: /etc/default/grub.d/99-mlperf-mem-cap.cfg
    permissions: '0644'
    content: |
      GRUB_CMDLINE_LINUX_DEFAULT="${GRUB_CMDLINE_LINUX_DEFAULT} mem=32G"
runcmd:
  - update-grub
```

After applying such GRUB configuration
`$free -h` on clients shows available `Mem: 29Gi`

- Quantity: 32 nodes (client fleet size for the CLOSED training submission
- Chassis: Nebius Cloud VM (virtualized; no discrete physical chassis exposed to the tenant)
- CPU: AMD EPYC-Genoa Processor, 1 socket, 32 cores, 64 vcpu
- Memory: 32 GiB per node limited by GRUB config
- Operating system: Ubuntu 24.04
- Networking: Ethernet, 1 virtual interface per node